GLITCHCON 2.0

VIT CHENNAI | 15 | ECDS | VITAA

NEST - Networked Escrow for Secure Transactions

Your funds, safely nested.

TEAM NAME: Runtime

GLITCHCON 2.0

VIT CHENNAI | 15 | ECDS | VITAA

GUIDELINES

- GitHub Submission: Submit your GitHub repository 8 hours before final evaluation.
- Deployment: Deploy the complete project repository (frontend deployment optional).
- Plagiarism Check: All submissions will undergo plagiarism verification.
- Pitch Format: Round 1 & Round 2 will include a 7-minute project pitch + evaluation.
- Team Requirement: Minimum 3 team members must be present during evaluation.

GLITCHCON 2.0

VIT CHENNAI 15 ECDS VITAA

PROBLEM STATEMENT

Infrastructure and transportation logistics often face **payment disputes, delayed settlements, and lack of transparency** between customers and suppliers. Traditional systems rely on **manual verification of delivery records**, leading to inefficiencies, mistrust, and payment delays.
Therefore, a **secure and automated payment mechanism** is needed to ensure payments are **released only after successful delivery confirmation**.

Objective:

Design and implement a **smart contract based escrow payment system** that:

- ❑ Locks payment securely when an order is placed
- ❑ Releases payment automatically after delivery confirmation
- ❑ Provides transparent and tamper proof payment tracking using blockchain
- ❑ This system aims to **improve trust, reduce disputes, and automate payment settlements** in transportation and infrastructure services.

GLITCHCON 2.0

VIT CHENNAI ECDS VITAA

OUR SOLUTION

- Develop a Blockchain based Smart Contract Escrow Payment System for transportation and logistics services.
- Customer payments are securely locked in a smart contract escrow when an order is placed.
- The driver completes the delivery and submits proof of delivery through the mobile application.
- The customer verifies the delivery using confirmation methods such as QR Delivery Confirmation.
- Once delivery is confirmed, the smart contract automatically releases payment to the driver or supplier.
- All transactions are recorded on blockchain for transparency and trust.

GLITCHCON 2.0

VIT CHENNAI

ECDS

VITAA

Our Features

- Escrow Smart Contract Payment
- Automatic Payment Release
- Blockchain Transaction Tracking
- Razorpay Payment Integration
- Smart Contract Delay Penalty
- Multi-Stage Payment Support
- AI Fraud Detection
- Driver & Supplier Reputation Score

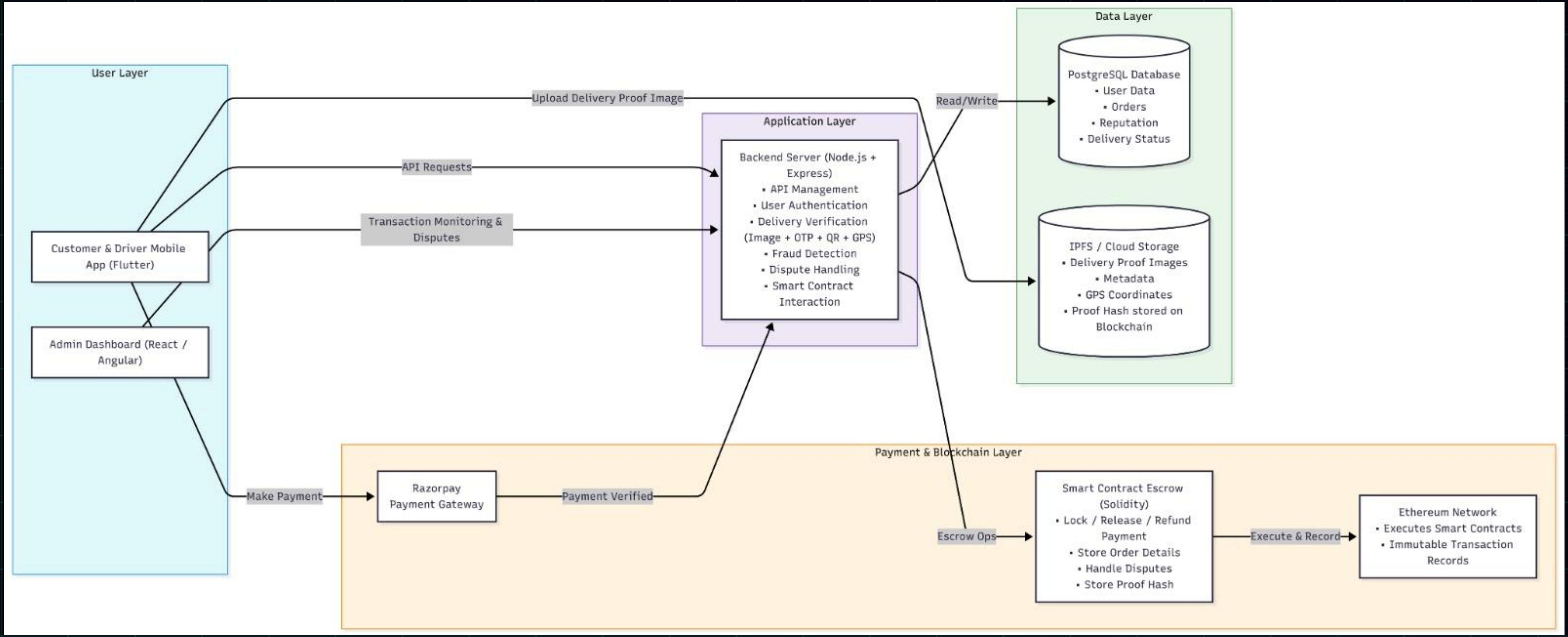
GLITCHCON 2.0

VIT CHENNAI | ECDS | VITAA

INNOVATION & USP

- Secure OTP / QR Delivery Confirmation – prevents fake delivery completion by verifying with customer OTP or QR code.
- Real-Time Delivery Tracking Interface – customers can track driver movement and delivery progress live.
- Escrow Transparency Dashboard – admin panel displaying blockchain transaction hashes and payment status.
- Address Confidence Score (AI) – predicts delivery success using address completeness, driver history, and location accuracy.
- Driver Feedback & Address Improvement System – drivers can report difficult locations to improve future deliveries.
- Smart Location Verification System – uses GPS pin drop and live location sharing to ensure accurate delivery locations.

WORKFLOW DIAGRAM WITH TECH STACK



GLITCHCON 2.0

VIT CHENNAI 15 ECDS VITAA

Tech Stack

Blockchain Layer

* Ethereum (Testnet – Sepolia)
* Solidity Smart Contracts
* Hardhat (Smart Contract Development)

Backend

* Node.js
* Express.js

Blockchain Integration

* Ethers.js / Web3.js

Mobile Application

* Flutter (Customer & Driver App)

Admin Dashboard

* Angular / React

Database (Off-chain Storage)

* PostgreSQL

File Storage

* IPFS / Cloud Storage (Delivery Proof Images)

Payment Gateway

* Razorpay

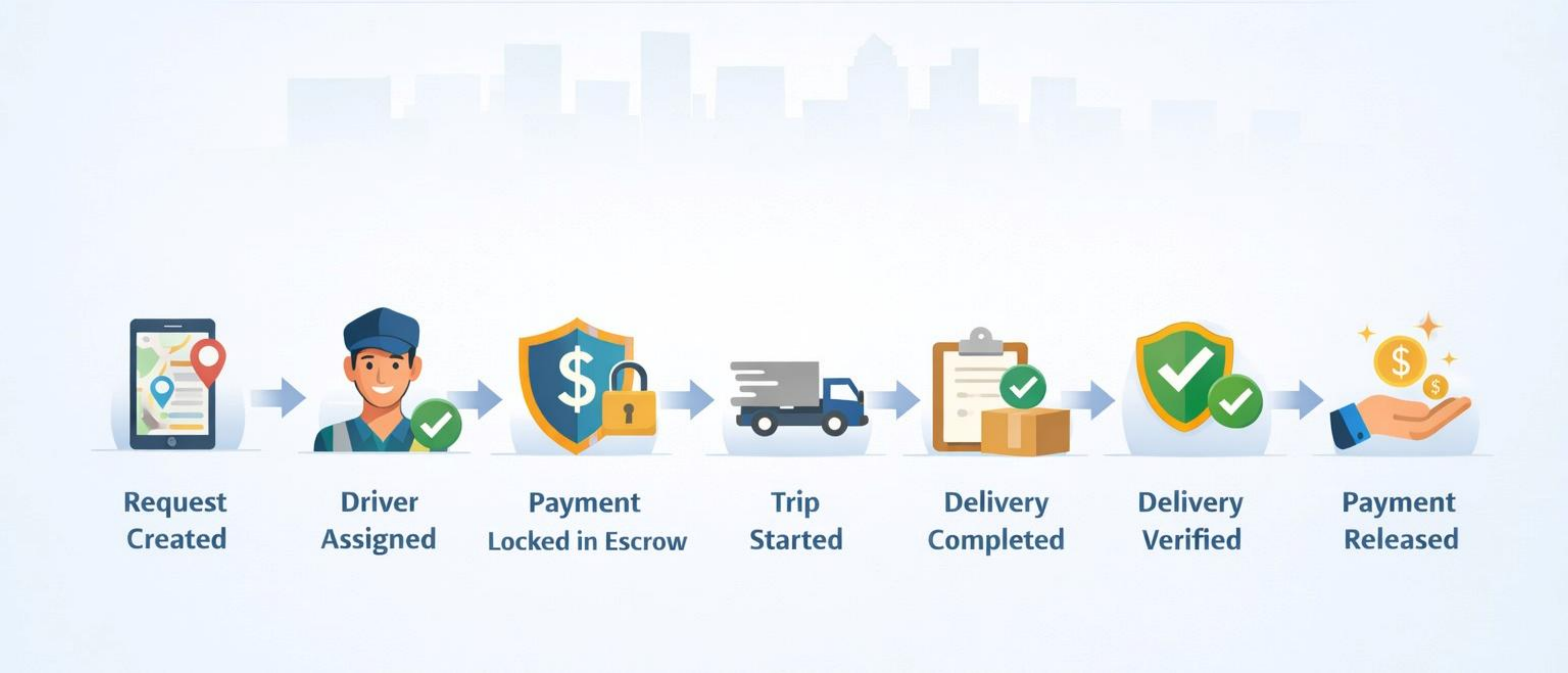
Authentication & Security

* JWT Authentication
* OTP Verification System

GLITCHCON 2.0

VIT CHENNAI ECDS

Lifecycle of a transportation order



GLITCHCON 2.0

VIT CHENNAI

ECDS

VITAA

TEAM MEMBERS

Manan Dudeja – 24BCE1270

Arnav Thapliyal – 24BCE1799

Akshat Singh – 24BCE1278

Tanushree Adhikari – 24BCE1439